Capstone — what to say on each slide

May 19, 2026 · 8:30 AM · UCC. Phone-readable. Each slide has bullets = the things to actually say, in order. Paraphrase, don't memorize.

Target: 24 min speaking + 6 min Q&A.

1 — Cover

- A few weeks ago I spent seven hours debugging a sensor before realizing the chip was shorted from the factory.
- The week before that, I'd finished the Google AI Professional Certificate.
- The gap between those two experiences is what I want to talk about.
- I'm Reuben Lavin, WPGA Class of 2026.
- My essential question: how do I transition from a consumer of AI to a developer through self-directed study?

[PAUSE 2s — look at the assessor before advancing]

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2 — Agenda

- Past, present, future, then questions.

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3 — Past (report cards)

- Some background on my learning habits — three lines from my report cards across high school.
- Grade 8: I avoided hard things. I played a lot of video games.
- Grade 10: I procrastinated and panicked at the end. Still gaming a lot, all the way into Grade 11.
- Grade 12: I'm running my own projects and meeting my own deadlines.

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4 — EQ + POA

- The question: how do I transition from a consumer of AI to a developer through self-directed study?
- The plan, four milestones:
- 01 knowledge baseline — the Google AI Pro Cert
- 02 pivot — Sentinel
- 03 integration — bullseye
- 04 synthesis — the lidar helmet
- What I'm about to show you is what actually happened against that plan.
- The deviation is where the learning is.

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5 — Phase 1 (Google AI Pro Cert)

- Phase one took the EQ literally.
- To become a developer through self-directed study, I went and got the credential.
- Feb 28 to Mar 5 — seven days.
- 11 Google Career Certificate courses.
- 8 Credly badges, including the Google AI Professional Certificate.
- It felt like sugarcoated grunt work.
- Resume padding more than real skills.
- Mostly corporate AI tools and consumer-level stuff.

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6 — "theory only gets you so far."

- Seven days of clicking through course modules taught me something.
- But it was passive.
- I was curious about the real thing.
- I wanted to be hands-on.
- To actually make something — instead of watching videos about other people making things.
- The next two months were that.

[PAUSE 2s before advancing]

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7 — Hardware lineage (text)

- Before any of the AI work, I had a hardware lineage.
- Fixed my old RC car. *(TODO: waiting on the right footage — green RC-tank-style car — from Reuben)*
- Built an autonomous LEGO car with ultrasonic obstacle avoidance.
- Built an RC airplane.
- Tore down and rebuilt an electric scooter.
- Hardware and software are one thing to me. The capstone work leans on this lineage.

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8 — Hardware gallery

- Top-left is the autonomous car — LEGO chassis, Arduino, ultrasonic.
- The looping clip is one of my plane test flights.
- The green PCB is the scooter.

[Advance after ~25s]

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9 — Sentinel (text)

- Sentinel — first project where I shipped a working AI integration end-to-end.
- Built in a hotel room during spring break in Mexico City.
- A camera feeds into a program that runs a person-detection AI.
- The AI is accelerated by an Intel chip.
- A live dashboard shows who's in the room and where they are.
- This was the project that told me I could actually do this.
- Bullseye and the helmet came out of the confidence Sentinel gave me.

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10 — Sentinel gallery

- That's the live AI output — boxes around the people the system has found, dots tracing how they're standing.
- I also tested AI skeletal tracking in this project, though it's not in this picture.

[Advance after ~15s]

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11 — Bullseye (text)

- My favorite story from the project — it shows the rhythm.
- **April 29** — version one, **salvage-radar**.
- A Craigslist scraper for free and cheap robotics parts near me.
- **May 2** — rebuilt as **bullseye**.
- Hit salvage-radar's limits 3 days in.
- Added an AI agent and price-comparison data.
- The legal piece:
- I researched it.
- Scraping Facebook Marketplace is in a grey area, but doable.
- Started from some abandoned GitHub repos for the scraping logic.
- Built my own utilities around them.
- **May 4** — rebuilt again as **bullseye-app**.
- A Windows desktop app.
- Scores Facebook Marketplace listings against actual eBay sold-prices.
- Open-source.
- 3 GitHub stars in the first week.
- Not a lot — but it's a work in progress.
- Still optimizing. Haven't started advertising yet.
- The lesson isn't the marketplace bot.
- It's that I shipped, used it, found what was broken, and rebuilt.
- Three times in five days.

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12 — Bullseye gallery

- Three views:
- landing page
- home dashboard with savings and streak
- a featured find — iPhone scored 75/100 against eBay sold comps
- Live at **getbullseye.app** — anyone in the room can visit it after this.

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13 — Helmet (text)

- The project this capstone was nominally about.
- An assistive-vision helmet for blind users.
- **Why:**
- my friend James built a vibrating cane for blind users
- I helped him wire it
- watching him test it, I started wondering what a helmet could pick up that a cane couldn't
- same problem, more spatial information
- **Works:**
- a chip in the helmet streams a depth map of the surroundings ~15 times a second to a laptop
- the laptop draws what the sensor sees in 3D, live
- **Stuck:**
- the motion sensor was broken from the factory
- returned it
- the part that tracks which way you're facing isn't running yet
- You can debug your code for hours — the answer is still that the chip is broken.
- **Lesson:** hardware quality is a variable you can't fix with skill.

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14 — Helmet gallery

- Top-left: the full setup running.
- Other shots:
- close-ups of the sensor
- second sensor under test
- multimeter session debugging the dead motion-sensor chip

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15 — Shorthand

- The project I struggled with most.
- **Idea:** type a two-letter code → AI expands it into a full prompt.
- I built the terminal version.
- Including the logic that picks the best expansion to suggest.
- Got it working.
- Then I realized I'd been polishing the wrong piece.
- The terminal version is one tiny corner.
- The real value lives where the shorthand works in *any* text field, anywhere on your computer.
- The deeper problem:
- I was optimizing decoding accuracy for sentences typed with way fewer characters.
- Across many samples, a plain Claude model with none of my optimization did just as well.
- My system added no benefit.
- Once I move to the system-wide version, the polish I just did is gone.
- **Lesson:** figure out where the real value lives before you start polishing.

[PAUSE 2s]

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16 — cc-discord-remote

- Two days ago I built **cc-discord-remote**.
- The official remote-control for my AI assistant didn't work for me.
- My phone account and my laptop account are different.
- So I built my own — a Discord bot.
- I send a message from anywhere.
- It types the message into the AI on my laptop.
- The response comes back to me in Discord.
- Two-day build.
- **Lesson:** when a tool doesn't fit your setup, build the bridge yourself.

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17 — Web design (and this deck)

- In parallel with the embedded work, I've been writing my own web design course — **Designing with Claude Code**.
- The method is a loop:
- write the course
- take the course on a real project
- find what's wrong
- rewrite the course
- v1 and v2 are archived in place so anyone can watch the method develop.
- v3 is live.
- The deck you're looking at uses the v3 method.

18 — Portfolio (16 repos · 3 weeks)

- This is the body of work — sixteen repositories created during the capstone window, most after April 24.
- Embedded systems, computer vision, marketplace algorithms, AI-assistant tooling, low-level automation.
- The breadth is the point.

19 — Network

- Two informal conversations with practicing experts.
- **Dr. Yang** — Professor of Civil Engineering at UBC, leading a research lab in construction automation and building affordability. He looked at my projects and connected them to his lab's work:
- automation in construction
- building affordability
- drones using stereo vision and computer vision to autonomously inspect the safety of buildings and bridges
- That real-world handle is something coursework couldn't have given me.
- **Rhys Rustad-Elliott** — systems software engineer.
- Undergrad with distinction from the University of Toronto.
- Master's *cum laude* from VU Amsterdam in computer security.
- Years at Google and Elastic working on low-level Linux systems and runtime security.
- We talked about:
- his work ethic and curiosity
- the wide range of computer science projects he's done
- how he picks what to work on next

20 — "build what isn't in the syllabus."

- Coming back to the question.
- Self-directed AI learning isn't about credentials.
- It's about choosing problems that force you into territory the courses don't cover.
- The Pro Cert was a starting point.
- The portfolio was the answer.

[PAUSE 3s — stand still, look at the assessor]

21 — What's next

- Biggest first.
- Train my own AI model on USGS seismic data — the original EQ, full-circle.
- Continue the lidar helmet — new motion sensor, then full sensor fusion.
- Structured courses on the side: AWS Cloud Practitioner, Anthropic's AI courses, Coursera SQL.
- Launch bullseye properly — not three GitHub stars, actual users.
- Red-teaming work on openclaw.

22 — "still building."

- The original EQ wasn't wrong.
- It was premature.
- I needed three months of messy applied work before I could even attempt it properly.
- The questions keep coming.

[PAUSE 4s — hold the silence. Then advance.]

23 — Outtakes / build

- Timelapse of me building the RC plane.
- A second flight test of that same plane.
- Brushless motor + propeller spinning on a bench test.
- The brushless motor itself, in my hand.

[~10s, then advance]

24 — Outtakes / AI tooling

- **claude-monitor** — a project I built to track my own AI usage on a small screen on my desk.
- The live dashboard running on the Pi.
- The "touch grass" nudge — it yells at me when I cross a usage budget.
- The Pi itself in a wooden case.
- Dashboard screenshot.

[~10s, then advance]

25 — Outtakes / bench

- Other projects + tinkering not on the main slides:
- ADC breakout bring-up on a breadboard
- SMD chip on my fingertip (scale shot)
- e-bike BMS controller PCB
- drone quad-copter assembly
- espresso machine internals I repaired
- lawnmower controller PCB
- first test drive of the RC car

[~10s, then advance to thank you]

26 — Thank you

- "Thank you. Questions?"
- Links shown on slide:
- github.com/reubenlavin08
- linkedin.com/in/reuben-lavin-4bb2712ba
- credly.com/users/reuben-lavin

Q&A — likely questions

- **What was hardest?** → Shorthand. I built the wrong layer before testing what users actually wanted.
- **What surprised you most?** → How much of self-directed AI work is choosing the right problem, not the right technique.
- **Would you do anything differently?** → Pre-arrange expert outreach. Two informal conversations should have been four or five structured ones with written questions.
- **Helmet roadmap?** → Replace the motion sensor → get pose tracking working → multi-sensor calibration → audio feedback for the user through bone-conduction headphones.
- **Why open-source bullseye?** → AGPL keeps it user-aligned, not enterprise-aligned. Anyone can fork it; commercial use requires the same license.
- **How did you decide what to build?** → Driven by my own friction. Bullseye started because I was hunting for free salvage parts. Helmet started because a friend built a cane.
- **What about the mentor meetings?** → I had one early on. My mentor read me as someone who didn't need much help and said so, so I ran autonomously.

Backup phrases if you blank

- "Let me come back to that."
- "I'll show you in a second."
- "The short version is..."
- "I don't have a clean answer to that — what I can say is..."